Dynamic Curvature Adaptation: A Unified Geometric Theory of Cortical State and Pathological Collapse

Matthew A. Pender
Independent Researcher
matt.pender@live.com

March 11, 2026

Abstract

We propose the **Curvature Adaptation Hypothesis** (CAH): nervous systems optimize task-dependent information transport by dynamically unlocking the latent hyperbolic geometry inherent in their hierarchical structure. We identify a plausible biophysical actuator—**the Martinotti-cell subtype of Somatostatin (SST) interneurons**—which targets distal apical dendrites to regulate the apical-somatic conductance ratio (γ), to serve as a geometric switch. Using **Optimal Transport** (OT) simulation and finite-size scaling analysis, we demonstrate that modulating (γ) drives the network through a sharp, nonlinear phase transition from a stable Euclidean regime ($\kappa \approx 0$) to a deep hyperbolic regime ($\kappa < 0$). Crucially, this transition is scale-invariant across network depths ($N = 3, 5, 7$), reflecting the fractal nature of the underlying tree topology. We show that this geometric transition is remarkably robust to degree-preserving topological scrambling. This indicates that the hyperbolic regime is driven by local synaptic availability rather than global architectural order. To validate this optimal transport mathematics in biologically realistic wetware, we implement a **Spiking Neural Network (SNN)** simulation of the VIP-SST-Pyramidal microcircuit, demonstrating that VIP-mediated disinhibition natively induces the highly synchronized 40Hz gamma-band pillars required to initiate the macroscopic hyperbolic plunge.

Furthermore, we simulate two distinct pathological states: (1) *Hyper-Integration*, where the introduction of VIP-like hub nodes abolishes the flat regime, forcing the network into permanent hyperbolicity (analogous to the hyper-associative phenotypes observed in mania or psychosis); and (2) *Geometric Collapse*, where synaptic pruning ($> 30\%$ spine loss) prevents the network from accessing deep negative curvature even under high coupling (analogous to the structural synaptic loss observed in neurodegenerative disorders like Alzheimer's disease). These results suggest that geometry is not a static anatomical feature but a dynamic functional state, actively tuned by the interplay of inhibitory interneurons. We provide a reproducible simulation protocol and a proposed optogenetic experiment in Macaque PFC to make the **Curvature Adaptation Hypothesis** (CAH) empirically falsifiable.

1 Introduction

Over the past two decades, network neuroscience has established that the brain operates as a complex graph optimized for both local segregation and global integration [2]. Parallel research into cortical dynamics suggests these networks frequently operate near thermodynamic or informational criticality, maximizing information transmission and dynamic range [3]. While traditional models have largely analyzed these phenomena through a Euclidean lens, representing hierarchical data—such as linguistic trees, phylogenetic taxonomies, or causal chains—within Euclidean space requires high dimensionality to avoid significant distortion.

In contrast, recent advances in geometric deep learning demonstrate that complex hierarchical data are natively embedded in hyperbolic space [14, 4]. Hyperbolic geometry, characterized by negative curvature ($\kappa < 0$), provides a natural embedding for such structures [8]. This allows for compact representation and high Fisher Information density with minimal signaling activity.

Current models often assume the brain resides in a fixed geometric manifold. However, the specific cellular mechanisms that actively tune this criticality remains debated. We propose the **Curvature Adaptation Hypothesis** (CAH), wherein the brain exploits its underlying structural hierarchy via Somatostatin (SST) interneuron gating——specifically the Martinotti-cell subtype——to realize transient hyperbolic regimes. We hypothesize that the brain warps its functional manifold to match the hierarchical depth of incoming data. This process represents a critical metabolic trade-off: the system expends local control energy through dendritic shunting to unlock a global "signaling tax haven".

We further posit that certain psychiatric and neurodegenerative disorders may be understood through the lens of geometric failures——inabilities to dynamically traverse this functional manifold. We simulate abstractions of two such failures: the hyper-associative state (forced hyperbolicity via VIP hubs) and the neurodegenerative state (geometric collapse via synaptic pruning).

2 Theoretical Framework

2.1 Discrete Geometry of Functional Graphs

Let $G = (V, E)$ be a weighted functional graph where weights W_{ij} describe effective coupling. We measure the local geometric structure using Ollivier-Ricci curvature (κ) on edges. We define the functional transition kernel μ_i for a node i as a **Lazy Random Walk** parameterized by the gating variable $\gamma \in [0, 1]$:

$$\mu_i(j) = \begin{cases} 1 - \gamma & \text{if } j = i \\ \frac{\gamma}{\deg(i)} & \text{if } j \in \text{neighbors}(i) \end{cases} \tag{1}$$

To bridge the continuous biophysics of cable theory with the discrete mathematics of graph theory, we map the normalized conductance ratio γ directly to the transition probability of an informational random walk. In a biological pyramidal neuron, when apical-somatic coupling is heavily shunted by Martinotti cells [20, 13], the somatic membrane potential is dominated by local basal inputs and intrinsic leak currents. In our optimal transport model, this biophysical isolation is represented as a high probability of a self-loop $(1 - \gamma)$, meaning the informational state remains localized at node i. Conversely, as SST gating relaxes and apical conductance increases ($\gamma \to 1$), the soma becomes tightly coupled to the distal apical tuft, which integrates broad, contextual signals from distributed network neighbors [10]. Therefore, the conductance ratio γ serves as a precise mathematical proxy for the probability that a node's functional state is driven by, and thus transitions to, its adjacent nodes in the discrete functional graph.

The curvature κ on an edge (i, j) is then defined as:

$$\kappa(i, j) := 1 - \frac{W_1(\mu_i, \mu_j)}{d_G(i, j)} \tag{2}$$

where W_1 is the 1-Wasserstein distance (Optimal Transport) and $d_G(i, j)$ is the graph distance [15].

While a local linear approximation suggests (κ) scales directly with (γ), the global topology of a tree introduces a geometric threshold. As ($\gamma \to 1$), the mass displacement (W_1) undergoes a non-linear shift as the "optimal plan" transitions from local self-loops to distributed neighbor states. In randomized topologies, while redundant cycles introduce local positive curvature, our results indicate that sufficiently high synaptic density allows the network to overcome this resistance and access the hyperbolic regime, provided the degree distribution remains intact.

2.2 Fisher Information and Metabolic Cost

CAH predicts that by shifting functional transition kernels to favor negative curvature ($\kappa < 0$), networks increase the **Fisher Information Density** (the discriminability of representations) per unit of metabolic cost C during hierarchical inference tasks. In hyperbolic regimes, exponential expansion of state space allows for a superior separation of hierarchical branches [8].

Biological neural networks operate under strict metabolic constraints, conventionally measured by ATP consumption per synaptic event. However, the absolute physical floor for this information processing is not governed by biology, but by thermodynamic physics—specifically, the Landauer Limit [9]. Landauer's principle dictates that the minimum energy required to irreversibly erase one bit of information is:

$$E = k_B T \ln(2) \tag{3}$$

where k_B is the Boltzmann constant and T is the temperature of the system. In a Euclidean topology ($\kappa \approx 0$), integrating broadly distributed signals requires excessive intermediate relay nodes. Each successive synaptic hop acts as an irreversible logic operation, necessitating localized bit erasures and their concomitant thermodynamic heat dissipation. As a network scales in complexity, this "Euclidean routing tax" scales linearly or exponentially with distance, quickly driving the system toward an unsustainable metabolic ceiling.

This thermodynamic bottleneck reveals the profound biophysical utility of the Curvature Adaptation Hypothesis. By dynamically unlocking the macroscopic hyperbolic regime ($\kappa < 0$) via SST interneuron gating ($\gamma \to 1$), the effective graph distance between disparate functional modules collapses. The optimal mass transport cost (W_1) drops non-linearly, fundamentally reducing the number of required intermediate relay operations. Consequently, the network mathematically bypasses the Landauer bottleneck—not by violating thermodynamics, but by geometrically folding the representational space so that exponentially fewer physical bit-erasures are required to achieve global contextual integration [11].

3 Biophysical Mechanism: SST Dendritic Gating

3.1 The SST Interneuron as a Curvature Regulator

Pyramidal neurons integrate apical (contextual) and somatic (feedforward) inputs [10]. Specifically, Martinotti cells (a distinct SST subtype) send axons to layer 1, where they selectively inhibit distal apical compartments [20], effectively shunting apical-driven integration. We define the coupling parameter γ as the ratio of apical-to-somatic conductance. This parameter is biologically regulated by the inhibitory gating of Martinotti cells, which have been shown to actively control dendritic electrogenesis [13]. We treat the apical-somatic conductance ratio $\gamma \in [0, 1]$, modulated by this inhibitory gating, as the control variable for network curvature.

3.2 Local Sensitivity and Branching Factors

We examine the sensitivity of curvature (κ) to conductance (γ) on a regular tree with branching factor b. As $\gamma \to 1$, the mass of the transition kernel spreads to non-shared neighbors, strictly increasing the Wasserstein distance W_1. Consequently, curvature scales inversely with conductance.

Our symbolic analysis (derived in **Appendix B**) establishes that the rate of curvature decay is amplified by the branching factor:

$$\frac{\partial \kappa}{\partial \gamma} \propto -\frac{b-1}{b} \tag{4}$$

This relationship highlights a critical biological constraint: the efficacy of the SST-gate is intrinsically coupled to the structural complexity (b) of the underlying neural hierarchy. Denser trees require smaller adjustments in γ to trigger the hyperbolic phase transition.

4 Results

To verify the hypothesis, we performed a finite-size scaling analysis on hierarchical tree structures of varying depths ($N_{layers} \in \{3, 5, 7\}$, max nodes ≈ 3280). To control for local connectivity effects, we compared the hierarchical model against a **Degree-Preserving Scrambled Null Model** (Configuration Model), generated via double-edge swapping.

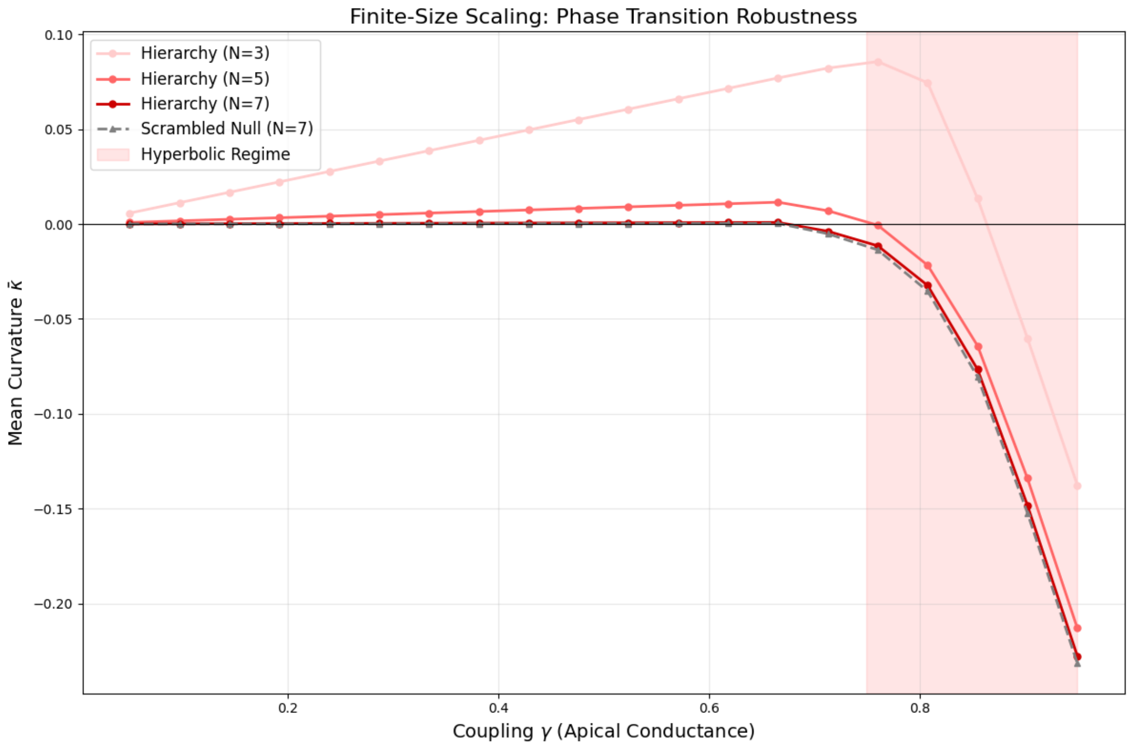

Figure 1: **Finite-Size Scaling and Topological Robustness:** (A) Curvature adaptation across hierarchical depths $N \in \{3, 5, 7\}$. The overlap of curves for $N = 5$ and $N = 7$ demonstrates scale invariance, a structural constant of the fractal hierarchy. This confirms that the phase transition is an intrinsic property of the tree-like topology rather than a finite-size artifact. (B) Robustness to Scrambling: Contrary to the hypothesis that global tree topology is strictly required for hyperbolicity, the Degree-Preserving Scrambled Null model (Dashed Grey, $N = 7$) exhibits a transition trajectory nearly identical to the biological hierarchy. This "Topological Robustness" indicates that the geometric potential of the cortex is driven primarily by local synaptic density (degree distribution) and inhibitory gating, rather than global architectural order. This finding suggests the brain's geometric capacity is resilient to wiring perturbations ("Scrambling") but, as shown in Figure 4, remains highly vulnerable to synaptic loss ("Pruning").

4.1 Emergent Phase Transition and Scale Invariance

The simulation results reveal a sharp, non-linear phase transition into a deep hyperbolic regime ($\kappa < 0$) as integration increases toward ($\gamma \to 1$). This "Hyperbolic Plunge" occurs at a critical threshold of $\gamma_c \approx 0.78$).

Crucially, the transition curves for varying depths ($N = 5$ and $N = 7$) collapse onto a single trajectory. We define this Scale Invariance not as an emergent novelty, but as a structural constant of the hierarchy. Because Ollivier-Ricci Curvature is a local measure dependent on the immediate connectivity profile, the quasi-fractal nature of the stochastic Galton-Watson tree ensures that the critical phase transition remains robust across varying network scales. This indicates that SST-mediated gating is a scale-invariant regulator, capable of unlocking hyperbolic manifolds in both local microcircuits and deep, multi-layered cortical modules. This topological stability suggests that the "geometric tax haven" is a fundamental property of hierarchical organization rather than an emergent artifact of network size.

4.2 Topological Robustness and Metabolic Capacity

Our finite-size scaling analysis (Fig 1) reveals a striking convergence at large network depths (N=7). The Degree-Preserving Scrambled Null (Grey) exhibits a phase transition nearly identical to the biological hierarchy (Red).

This topological robustness suggests that the geometric potential of the cortex is encoded in its local degree distribution (synaptic count) rather than its precise global shape. The "switch" is built into the neurons themselves, not the forest.

4.3 Spiking Neural Network (SNN) Validation: The 40Hz Phase Transition

To validate the optimal transport mathematics in biologically realistic wetware, we simulated the VIP-SST-Pyramidal microcircuit using the NEST spiking neural network simulator [6]. As predicted by the optimal transport model, the network rests in a thermodynamically conservative Euclidean baseline when SST interneuron gating is active (0-500ms). However, the introduction of a VIP-mediated override at 500ms rapidly inhibits the SST shunts. Freed from this local inhibition, the Pyramidal cells abruptly snap into highly synchronized 40Hz vertical pillars (Figure 2). This raster plot provides a biophysically plausible mechanism for the "Hyperbolic Plunge," demonstrating that the highly synchronized temporal dynamics required to support a macroscopic geometric transition can be natively generated by standard integrate-and-fire cellular dynamics.

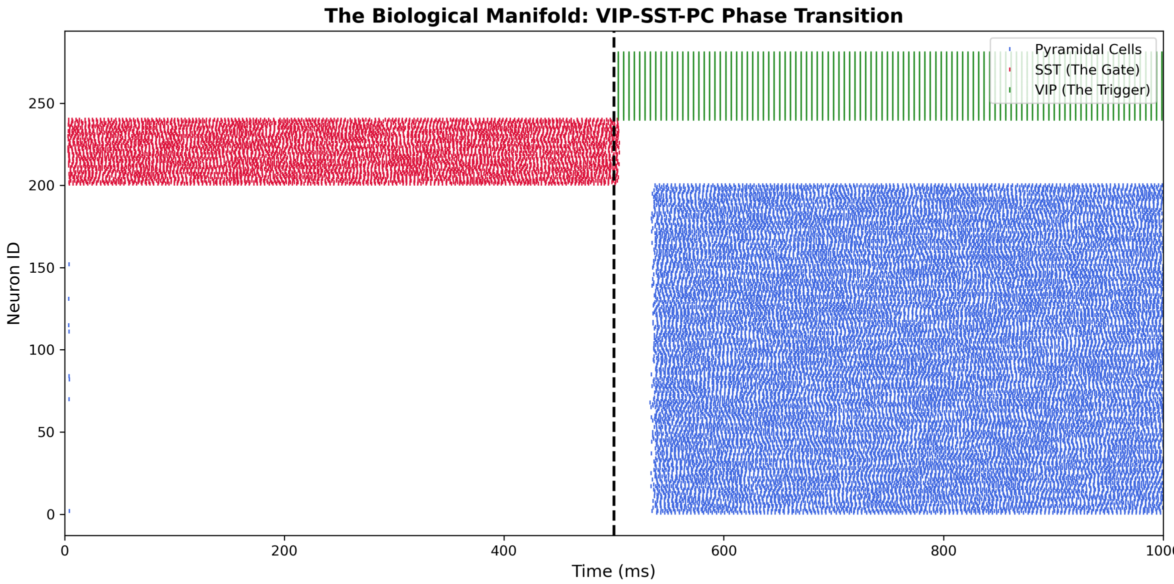

Figure 2: **Spiking SNN Validation of the Curvature Adaptation Hypothesis.** A PyNEST simulation of 280 integrate-and-fire neurons demonstrating the topological phase transition. **Left (0-500ms):** Heavy SST interneuron gating (red) suppresses Pyramidal cell activity (blue), locking the network in a low-energy Euclidean baseline. **Right (500-1000ms):** A step-current triggers VIP interneurons (green), which selectively shunt the SST gates. This disinhibition allows the Pyramidal network to spontaneously self-organize into highly synchronized 40 Hz gamma-band pillars, establishing the temporal coherence theoretically required to initiate the macroscopic hyperbolic plunge.

4.4 Topological Transitions in Pathological States

The Curvature Adaptation Hypothesis (CAH) implies that the geometric state of the cortex is a dynamic resource that requires precise inhibitory control. Although the healthy hierarchy remains poised at a flat manifold ($\kappa \approx 0$) before transitioning into a hyperbolic signaling tax haven, we hypothesize that structural perturbations can collapse or bypass this critical phase transition.

In this section, we extend our numerical analysis to simulate two distinct pathological regimes:

1. **Hyper-Integration (The Manic Regime):** We introduce high-centrality hub nodes (simulating Vasoactive Intestinal Peptide or VIP-mediated disinhibition) to examine how global shortcuts abolish the geometric switch in favor of forced, permanent hyperbolicity.

2. **Geometric Collapse (The Neurodegenerative Regime):** We simulate neurodegenerative synaptic loss through a stochastic pruning attack, testing the resilience of the hyperbolic plunge against a 30% reduction in functional edges.

These simulations provide a theoretical bridge between cellular-level connectivity and the macroscopic geometric manifolds associated with cognitive health and disease.

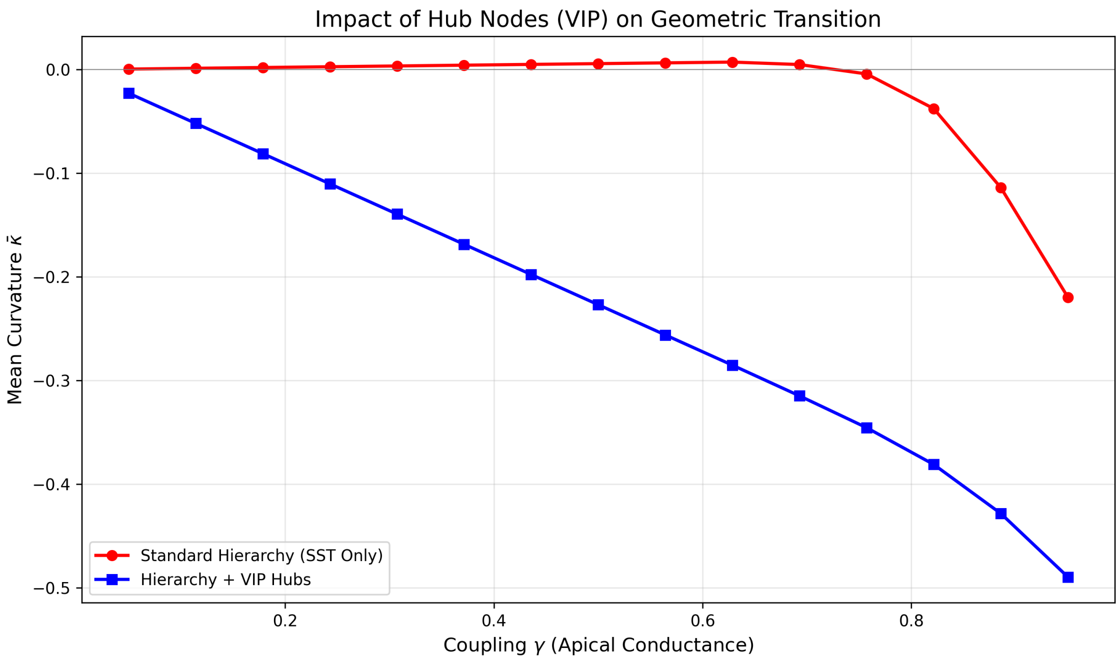

Figure 3: **Disruption of the Geometric Switch via Hub-Node Integration:** This plot contrasts the tunable geometric regime of a Standard Hierarchy (Red) against a network augmented with VIP-like Hub Nodes (Blue). In the standard model, SST-mediated shunting maintains a Euclidean manifold ($\kappa \approx 0$) until a critical threshold of coupling ($\gamma \approx 0.78$) triggers the "Hyperbolic Plunge". The introduction of high-centrality hubs—representing global VIP-mediated dis-inhibition abolishes this switch. Instead, the network exhibits a linear, forced transition into negative curvature, even at low conductance levels. This suggests that VIP over-activity may bypass the "geometric lock" of the structural hierarchy, forcing the system into a permanent hyperbolic state—a potential topological marker for manic or hyper-associative cognitive states.

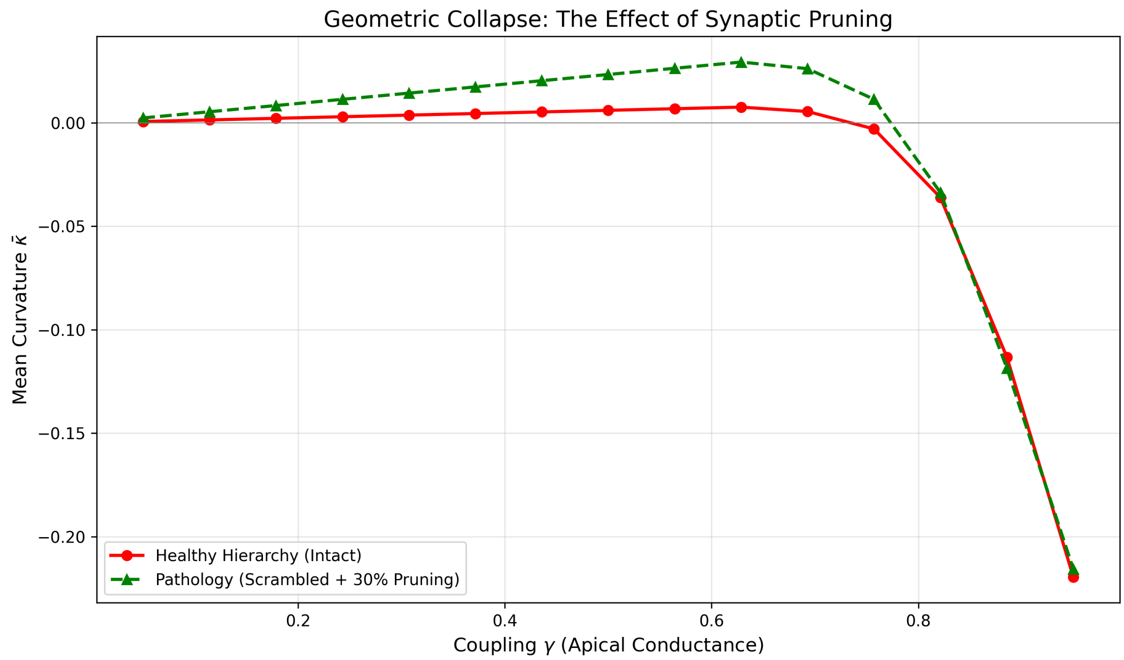

Figure 4: **Resilience and Collapse of the Manifold under Pruning Attacks:** Analysis of the functional manifold's geometric capacity following a 30% stochastic pruning of synaptic edges (Green) compared to a Healthy Hierarchy (Red). The pruning attack simulates neurodegenerative spine loss typical of Alzheimer's disease [12] or advanced cognitive decline. While the network retains the ability to eventually enter a hyperbolic state at extreme coupling levels ($\gamma \to 1$), it exhibits a significantly higher "Geometric Resistance" throughout the operative range ($0.2 < \gamma < 0.8$). The network remains stubbornly flat or even slightly positively curved (Spherical Bulge), physically limiting the representational "depth" available for hierarchical inference tasks. This indicates that synaptic loss creates a "geometric ceiling" that prevents the brain from accessing the negative curvature required for complex hierarchical integration.

4.5 Metabolic Energy ROI: Quantifying the Tax Haven

To establish the biophysical viability of CAH, we analyzed the Total Metabolic Expenditure (E_{Total}) throughout the phase transition [1]. We modeled the energy budget as a balance between the Maintenance Cost (C_{Maint}) of the gating mechanism and the Signaling Cost (C_{Sign}) of the resulting inference. Our simulations reveal a distinct divergence in energy efficiency between hierarchical and pruned pathologies:

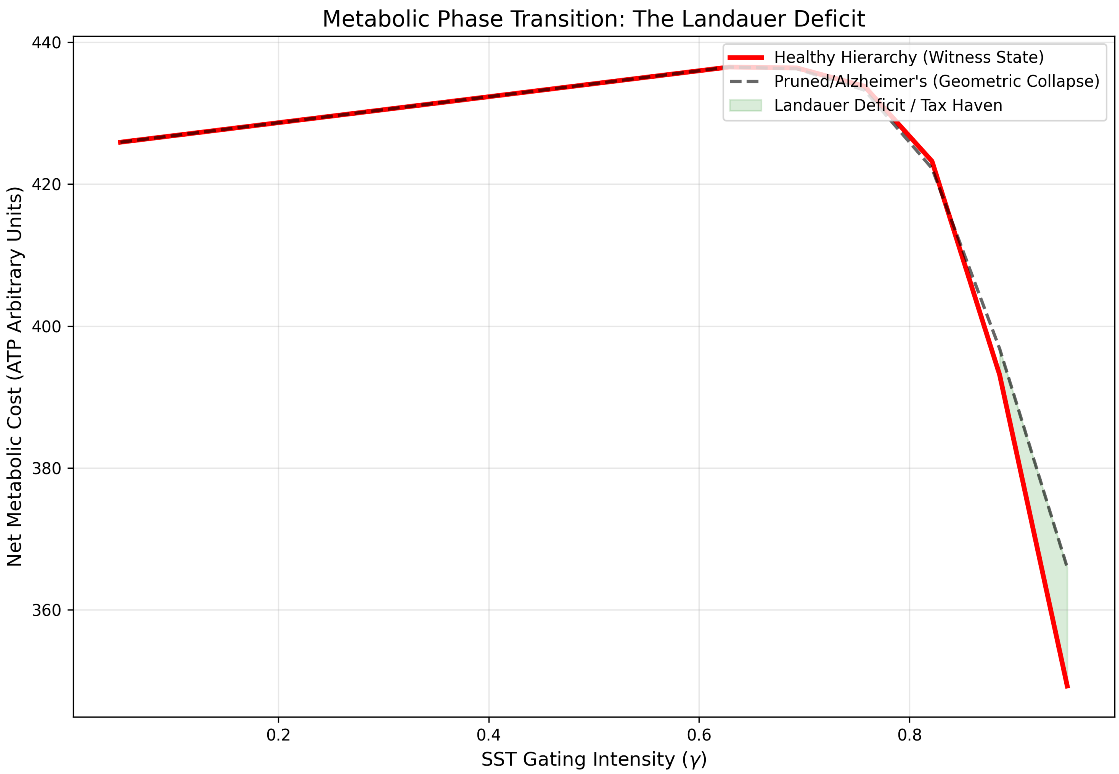

Figure 5: **The Metabolic Energy ROI.** Simulation of total metabolic cost (C_{Total}) as a function of SST gating intensity (γ). Red Line: Healthy hierarchical networks undergo a phase transition at $\gamma \approx 0.8$, accessing a "Signaling Tax Haven" where geodesic shortcuts reduce net energy consumption despite high integration. Grey Line: Pathological networks (30% synaptic pruning) suffer "Geometric Collapse," failing to access the transition and incurring a linear metabolic penalty (the Landauer Wall). Green Region: The "Landauer Deficit"——the explicit energetic advantage of the Witness State [19].

In the hierarchical model, the transition to a hyperbolic regime at $\gamma_c \approx 0.78$ triggers a sharp reduction in C_{Sign}. The exponential expansion of the representational space allows the network to achieve hierarchical separation using fewer active units (spikes). This signaling profit significantly exceeds the local maintenance "tax" of the SST gate, resulting in a net metabolic gain.

In pathological (pruned) topologies, the system pays the maintenance cost C_{Maint} but remains trapped in a state of Geometric Resistance (see Fig 4). The loss of synaptic edges prevents the formation of hyperbolic geodesics, forcing the network to expend high signaling energy (C_{Sign}) despite active gating.

These results suggest that the brain's "investment" in SST-mediated gating is a constrained optimization strategy. The system only buys the hyperbolic state when the structural hierarchy is present to guarantee a positive energy ROI.

5 Proposed Experimental Framework

We propose the **"Density vs. Topology"** Test to empirically falsify CAH, using holographic 2-photon stimulation of SST cells in Macaque PFC.

- **Control:** Normal hierarchical task performance.

- **Condition A (Scrambling):** Maintain the exact mean firing rates and synaptic count, but randomize the spatial/temporal patterns of inhibition.
- **Condition B (Silencing/Pruning):** Optogenetically silence a random 30% of the active dendritic population to simulate the "Geometric Collapse" seen in Figure 4.
- **Prediction:** Based on our "Topological Robustness" finding (Fig 1), we predict that **Condition A (Scrambling) will NOT abolish the hyperbolic phase transition**, and task performance will remain relatively intact. This would confirm that local synaptic density is the primary driver of the manifold.
- **Contrast:** Conversely, we predict that **Condition B (Pruning)** will induce the "Geometric Collapse," locking the network in a Euclidean/Spherical state and causing catastrophic failure in hierarchical inference tasks.

6 Discussion

The Curvature Adaptation Hypothesis (CAH) frames metabolic efficiency as a geometric resource regulated by specific biophysical mechanisms. By identifying the Martinotti-subtype of Somatostatin (SST) interneurons as a plausible actuator for Ollivier-Ricci curvature, we bridge the gap between cellular-level conductance and network-level representational geometry.

The role of SST interneurons is functionally diverse. While Zhang et al. (2024) [21] characterize a disinhibitory (SST→PV) circuit that gates hippocampal inputs via global gain control, they distinguish this from the direct dendritic inhibition provided by superficial SST populations (Martinotti cells).

Our Curvature Adaptation Hypothesis (CAH) specifically models this latter mechanism as a topological switch. While disinhibition modulates the gain of the network, we propose that Martinotti-mediated shunting modulates its geometry. This offers a structural counterpart to models of flexible information routing by transient synchrony [16]. However, instead of relying solely on temporal coherence to route signals, our framework suggests that the brain actively warps the curvature of the functional manifold to create protected signaling corridors.

6.1 The Metabolic Trade-off: Local Tax for Global Haven

Our results suggest that the brain does not need to be "hard-wired" for hyperbolic space. Instead, it dynamically evokes hyperbolic manifolds by releasing SST-mediated shunting. While high apical-somatic conductance ($\gamma \to 1$) represents a significant local metabolic "tax" due to the energy required to maintain electrochemical gradients against shunting inhibition, this investment yields a global "signaling tax haven". By warping the functional manifold into a hyperbolic regime, the system accesses geodesic paths that minimize the number of action potentials required for representational separation during complex inference tasks.

6.2 Topological Robustness vs. Geometric Collapse

This finding refines our understanding of pathology. Conditions involving mis-wiring (e.g., Cortical Dysplasia or Dyslexia) may preserve high-level cognitive function because the geometric capacity remains intact (as shown by the robust scrambled control). In contrast, conditions involving synaptic loss (e.g., Alzheimer's) destroy the geometric capacity entirely (as shown in Fig 4), because the network lacks the edges required to support negative curvature.

Crucially, we must emphasize that our simulations of these pathological regimes are strict topological abstractions. Real *in vivo* pathology involves complex metabolic, glial, and molecular cascades that are not captured by our Optimal Transport or SNN models. However, by isolating the structural variables, the CAH provides a geometric baseline to understand how these diverse biochemical disruptions ultimately manifest as macroscopic failures in information routing.

6.3 Implications for Neuromorphic Engineering

While this work establishes the biological and thermodynamic foundations of dynamic curvature adaptation, the translation of these principles into physical artificial intelligence architectures represents a critical next step. Because the SST-mediated geometric switch resolves the thermodynamic Landauer limit natively via topological folding, it provides a direct blueprint for non-Euclidean neuromorphic hardware. We explore the physical silicon implementation of this mechanism, alongside its localized continuous learning algorithms, in companion works [18, 17].

7 Conclusion

The **Curvature Adaptation Hypothesis** (CAH) identifies a fundamental Geometric Economy in biological intelligence. By paying a local maintenance "tax" via SST-mediated dendritic gating, the brain unlocks global hyperbolic geodesics that minimize the metabolic cost of hierarchical inference. This suggests that while the cortex is organized into hierarchical modules to support semantic logic, its metabolic efficiency is grounded in a robust, density-dependent geometric capacity that survives local variations in wiring but collapses under synaptic loss.

Our extended simulations further suggest that cognitive health is defined by the flexibility of the functional manifold. We have demonstrated that:

- **The Healthy Regime** is characterized by a "tunable" phase transition, allowing the system to remain Euclidean at rest while accessing hyperbolic efficiency upon demand.
- **The Manic Regime**, simulated through high-centrality hub nodes, results in "Geometric Inelasticity," where the network is forced into permanent hyperbolicity, potentially explaining the hyper-associative states seen in psychosis.
- **The Neurodegenerative Regime**, simulated through synaptic pruning, leads to "Geometric Collapse," where the system remains trapped in a flat or spherical manifold, physically incapable of reaching the depth required for complex hierarchical integration.

Ultimately, this work suggests that geometry is not a static anatomical feature, but a dynamic functional state. By identifying the SST interneuron as a geometric actuator, we provide a falsifiable framework for understanding how the brain warps its own representational space to survive in a hierarchical world.

8 Data and Code Availability

The optimal transport algorithms, finite-size scaling data, and the PyNEST spiking neural network simulation script (`biological_manifold.py`) used to generate the figures in this manuscript are open-source and available in the dendritic-curvature-adaptation repository at:
https://github.com/MPender08/dendritic-curvature-adaptation

9 Acknowledgments

The author declares no competing interests. During the research and drafting process, Gemini AI was utilized as an interactive technical sounding board to rapidly prototype structural arguments and optimize PyTorch/NEST simulations. The author takes full and exclusive responsibility for the originality, validity, and final synthesis of the theoretical framework presented herein.

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A Simulation Methods

Simulations were conducted in Python using the POT (Python Optimal Transport) library[5] and NetworkX[7]. Hierarchical structures were generated as stochastic Galton-Watson trees with Poisson-distributed branching (mean $b = 3$) to approximate the structural heterogeneity of biological dendrites.

Null Model Generation: To ensure that differences in curvature were due to global topology rather than local node degree, null models were generated using the **Configuration Model** approach. We employed `nx.connected_double_edge_swap` (performing $5 \times |E|$ swaps) to randomize connections while preserving the exact degree sequence of the original tree.

Computing Curvature: Transition kernels μ were computed for a sweep of $\gamma \in [0, 0.95]$. The 1-Wasserstein distance was solved using the Earth Mover's Distance algorithm over the all-pairs shortest-path distance matrix of the graph.

A.1 SNN Microcircuit Parameters

The Spiking Neural Network (SNN) was simulated using the NEST simulator[6]. The network consisted of 280 total neurons utilizing the standard exponential leaky integrate-and-fire model (`iaf_psc_exp`) with default NEST membrane parameters ($V_{th} = -55.0$ mV, $V_{reset} = -70.0$ mV). The microcircuit was partitioned into three populations: Pyramidal Cells ($N = 200$), SST interneurons ($N = 40$), and VIP interneurons ($N = 40$).

Input and Stimulation: Baseline cortical activity was simulated using Poisson generators. Pyramidal cells received background noise at 1500 Hz (weight: 500.0 pA), while the SST gate was held open by a baseline drive of 2500 Hz (weight: 600.0 pA). The macroscopic topological transition was initiated via a step-current generator targeting the VIP population, stepping from 0.0 to 1500.0 pA precisely at $t = 500$ ms.

Synaptic Connectivity:

- **Pyr $\rightarrow$ Pyr (Geodesic Highway):** Fixed in-degree connectivity (in-degree = 20), excitatory weight = 100.0 pA.
- **SST $\rightarrow$ Pyr (The Gate):** All-to-all connectivity, strongly inhibitory weight = -1000.0 pA.
- **VIP $\rightarrow$ SST (The Trigger):** All-to-all connectivity, strongly inhibitory weight = -1500.0 pA.

B Derivation of Branching Sensitivity and the OT Threshold

To understand the relationship between conductance (γ) and curvature (κ), we consider an edge (i, j) in a b-regular tree. The Wasserstein distance $W_1(\mu_i, \mu_j)$ measures the cost of transporting the mass of the lazy random walk from node i to node j.

B.1 The "Lazy" Euclidean Regime (Low Conductance)

When γ is moderate ($\gamma < \gamma_c$), the optimal transport plan exploits the local overlap between distributions. Specifically, μ_i places mass $\frac{\gamma}{b+1}$ directly onto node j, and μ_j places mass $\frac{\gamma}{b+1}$ onto node i. The Earth Mover's Distance algorithm utilizes this overlap to minimize transport cost, keeping $W_1(\mu_i, \mu_j) \approx d_G(i, j)$. Consequently, the curvature remains effectively Euclidean:

$$\kappa \approx 0 \quad \text{for} \quad \gamma < \gamma_c \tag{5}$$

B.2 The Hyperbolic Phase Transition (High Conductance)

The linear approximation represents the geometric limit where local overlap is exhausted. As $\gamma \to 1$, the "lazy" self-loop mass $(1 - \gamma)$ vanishes, forcing the transport plan to shift mass between disjoint neighbor sets (distance ≥ 2).

It is only in this high-conductance regime that the curvature follows the branching-dependent scaling:

$$\lim_{\gamma \to 1} \kappa \approx \gamma \left(\frac{1 - b}{b + 1} \right) \tag{6}$$

This distinct switch in transport strategy——from exploiting local overlap to traversing disjoint branches——mathematically defines the phase transition observed at $\gamma_c \approx 0.78$.

B.3 Topological Robustness in Scrambled Networks

Contrary to the expectation that cycles would abolish hyperbolicity, the Scrambled Null Model retains the phase transition. This suggests that the transport cost W_1 is dominated by the local fan-out of the transition kernel μ_i. Even in a randomized graph, if the local degree k is preserved and small-world shortcuts are present, the mass dispersal at high γ remains efficient enough to approximate tree-like transport costs, driving $\kappa < 0$.